



Generation Interconnection

Impact Study Report

for

Queue Project AE2-258

CHASE CITY 115 KV

14.1 MW Capacity / 0 MW Energy

Revision 2, January 2022

Revision 1, December 2021

February, 2020

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1 Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (VEPCO).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 Revision 1 Summary- December 2021

This revision is being issued to incorporate results of a re-tool that was performed.

4 Revision 2 Summary- January 2022

The previous version identified network upgrade n6115 for a system reinforcement. It was identified that the scope of this work will be completed under s2614. Report updated to reflect s2614 as the required system reinforcement.

5 General

The Interconnection Customer (IC), has proposed an uprate to an existing solar generating facility located in Mecklenburg County, Virginia. This project requests a Capacity-only increase of 14.1 MW. The installed facilities will have a total capability of 49.9 MW with 33 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project 06/01/2020. This study does not imply a TO commitment to this in-service date.

Queue Number	AE2-258
Project Name	CHASE CITY 115 KV
Interconnection Customer	Bluestone Farm Solar
State	Virginia
County	Mecklenburg
Transmission Owner	Dominion
MFO	49.9
MWE	0
MWC	14.1
Fuel	Solar
Basecase Study Year	2022

5.1 Point of Interconnection

AE2-258 is an uprate to the AB2-043 queue position which will interconnect with the Dominion transmission system at the Chase City 115 kV substation.

5.2 Cost Summary

The AE2-258 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ N/A
Direct Connection Network Upgrade	\$ N/A
Non Direct Connection Network Upgrades	\$ N/A
Total Costs	\$ 0

In addition, the AE2-258 project may be responsible for a contribution to the following costs

Description	Total Cost
System Upgrades	\$ 0

Note: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start

with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

6 Transmission Owner Scope of Work

Dominion assessed the impact of the proposed Queue Project AE2-258 was evaluated as a 14.1 MW Capacity (0.0 MW energy) injection at the Chase City 115 kV substation in the Dominion Transmission System, for compliance with NERC Reliability Criteria on Dominion Transmission System. The system was assessed using the summer 2022 AE2 case provided to Dominion by PJM. When performing a generation analysis, Dominion's main analysis will be load flow study results under single contingency (both normal and stressed system conditions). Dominion Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of Dominion's Planning Criteria and interconnection requirements can be found in the Company's Facility Connection Requirements which are publicly available at: <http://www.dominionenergy.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically in Planning Studies NERC Category C Contingency Conditions (Bus Fault, Tower Line, N-1-1, and Stuck Breaker scenarios) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For Dominion Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

The required Attachment Facilities, Direct Connection and Non-Direct Connection work for the interconnection of the AE2-258 generation project to the Dominion Transmission System is detailed in the following sections. The associated one-line with the generation project attachment facilities and primary direct and non-direct connection are shown in Attachment 1.

Note that the ITO findings were made from a conceptual review of this project. A more detailed review of the connection facilities and their cost will be identified in a future study phases. Further note that the cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. ITO herein reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

7 Schedule

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimate elapsed time to complete each of the required reinforcements is identified in the “System Reinforcements” section of the report.

8 Transmission Owner Analysis

8.1 Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2022 summer peak load flow model and the results were verified by Dominion. Additionally, Dominion performed an analysis of its transmission system and no additional system reinforcements were identified.

8.2 Short Circuit Analysis

PJM performed a short circuit analysis and the results were verified by Dominion. The connection of AE2-258 project to the system does not result in any newly overdutied circuit breakers on the Dominion transmission system and does not have a significant fault current contribution to existing overdutied circuit breakers

8.3 Stability Analysis

Not required for this project.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion’s “Dominion Energy Electric Transmission Generator Interconnection Requirements” documented in Dominion’s Facility Interconnection Requirements “Exhibit C” located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC’s GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.

3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion’s “Dominion’s Facility Interconnection Requirements” document located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

9.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.1.1 Meteorological Data Reporting Requirement

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

10.2 Dominion Requirements

See Section 3.4.6 “Metering and Telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

11 Network Impacts

The Queue Project AE2-258 was evaluated as a 14.1 MW capacity uprate to an existing solar generating facility under AB2-043 in the Dominion area. The total capability of AE2-258 and AB2-043 is a MFO of 49.9 MW with 33.0 MW recognized by PJM as capacity. Project AE2-258 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-258 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

12 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

13 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

14 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
1	314562	6CLUBHSE	230	DVP	314563	6CLUBHSE	115	DVP	1	DVP_P4-6: CAROLIN T122	LFFB	209	124.41	125.37	AC	2.36
2	314562	6CLUBHSE	230	DVP	314563	6CLUBHSE	115	DVP	1	DVP_P4-2: 2202	LFFB	209	119.23	119.74	AC	2.22
3	314562	6CLUBHSE	230	DVP	314563	6CLUBHSE	115	DVP	1	DVP_P4-2: 101002	LFFB	209	118.52	119.03	AC	2.22
4	314562	6CLUBHSE	230	DVP	314563	6CLUBHSE	115	DVP	1	DVP_P4-2: 5402	LFFB	209	118.5	119.01	AC	2.22
5	314702	3KERR	115	DVP	304102	3GW KING TAP	115	CPEL	1	DVP_P7-1: LN 22-90-A	DCTL	199	131.64	133	AC	2.74
6	314702	3KERR	115	DVP	304102	3GW KING TAP	115	CPEL	1	DVP_P4-6: CAROLIN T122	LFFB	199	125.96	127.34	AC	2.77
7	314702	3KERR	115	DVP	304102	3GW KING TAP	115	CPEL	1	DVP_P7-1: LN 22-90-B	DCTL	199	120.48	121.85	AC	2.74

15 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

16 System Reinforcements

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AE2-258 Allocation
#1,2,3,4	6CLUBHSE 230/115 kV transformer	<u>Dominion</u> Add a second 230-115 kV transformer at Clubhouse substation to mitigate the existing transformer thermal overload. New Ratings: 276/292/328 MVA Time Estimate: 24-30 months Queue Project AE2-258 presently does not receive cost allocation for this upgrade. Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AE2-258 could receive cost allocation. Note 2: Although Queue Project AE2-258 may not have cost responsibility for this upgrade, Queue Project AE2-258 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-258 comes into service prior to completion of the upgrade, Queue Project AE2-258 will need an interim study.	n6114	\$6,000,000	\$0
#5,6,7	3KERR-3GW KING TAP 115 kV line	<u>Dominion:</u> Rebuild approximately 4.7 miles Line #45 between Kerr Dam to Duke Interconnection with current 115kV standards (768 ACSS which at 250 degree C). New Ratings: 393/393/452 MVA Project In-service: 12/31/2022 Note: Although Queue Project AE2-258 may not have cost responsibility for this upgrade, Queue Project AE2-258 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-258 comes into service prior to completion of the upgrade, Queue Project AE2-258 will need an interim study.	s2614	\$11,000,000	\$0
		<u>Duke/Progress Energy</u> For the Duke/Progress portion of the line, reconductor 20 miles with single 795 ACSS-TW per phase, replace breaker, disconnect switches, wave trap, line switches. Time Estimate: 48 months Note: The Duke/Progress Energy portion of this line is studied under Duke's FERC tariff process. Reference the latest applicable affected system study.	dep0002	\$60,000,000	\$0

17 Appendices

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the Appendices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the Appendices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators.

It should be noted the project/generator MW contributions presented in the body of the report and appendices sections are full contributions, whereas the loading percentages reported in the body of the report, take into consideration the commercial probability of each project as well as the ramping impact of "Adder" contributions.

17.1 Contingency Decryptions

Contingency Name	Contingency Definition
DVP_P4-2: 101002	CONTINGENCY 'DVP_P4-2: 101002' /* CAROLINA 115 KV
	OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00
	OPEN BRANCH FROM BUS 314259 TO BUS 314835 CKT 1 /* 3CAROL56_1 115.00 - 3CAROL_1 115.00
	OPEN BUS 314835 /* ISLAND: 3CAROL_1 115.00
	OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00
	OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00
	OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00
	OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00
	END

Contingency Name	Contingency Definition
DVP_P4-2: 2202	CONTINGENCY 'DVP_P4-2: 2202' /* CAROLINA 115 KV
	OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00
	OPEN BRANCH FROM BUS 314571 TO BUS 925780 CKT 1 /* 3EATON F 115.00 - AC1-054 TAP 115.00 /* CONTINGENCY LINE ADDED FOR AE1 BUILD
	OPEN BUS 314571 /* ISLAND: 3EATON F 115.00
	OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00
	OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00
	OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00
	OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00
	END
DVP_P4-2: 5402	CONTINGENCY 'DVP_P4-2: 5402' /* CAROLINA 115 KV
	OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00
	OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00
	OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00
	OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00
	OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00
	END

Contingency Name	Contingency Definition
DVP_P4-6: CAROLIN T122	CONTINGENCY 'DVP_P4-6: CAROLIN T122' /* CAROLINA 115 KV
	OPEN BRANCH FROM BUS 314559 TO BUS 315126 CKT 1 14.400 /* 3CAROLNA 115.00 - 1ROARAP2
	OPEN BRANCH FROM BUS 314559 TO BUS 315128 CKT 1 14.400 /* 3CAROLNA 115.00 - 1ROARAP4
	OPEN BUS 315126 /* ISLAND: 1ROARAP2 14.400
	OPEN BUS 315128 /* ISLAND: 1ROARAP4 14.400
	OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 115.00 /* 3CAROLNA 115.00 - 3EATON F
	OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 3CAROLNA 115.00 /* 3OCCONEECHEE115.00 -
	OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 3CAROLNA 115.00 /* 3CAROL56_1 115.00 -
	OPEN BRANCH FROM BUS 314559 TO BUS 314578 CKT 1 115.00 /* 3CAROLNA 115.00 - 3HORNRTN
	OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 115.00 /* 3CAROLNA 115.00 - 3L GASTN
	OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 115.00 /* 3CAROLNA 115.00 - 3PLHITP
	OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 230.00 /* 3CAROLNA 115.00 - 6CAROLNA
	OPEN BUS 314559 /* 3CAROLNA 115.00 KV
	END
DVP_P7-1: LN 22-90-A	CONTINGENCY 'DVP_P7-1: LN 22-90-A'
	OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 115.00 /* 3CAROLNA 115.00 - 3EATON F
	OPEN BRANCH FROM BUS 314571 TO BUS 925780 CKT 1 115.00 /* 3EATON F 115.00 - AC1-054 TAP
	OPEN BUS 314571 /* ISLAND: 3EATON F 115.00
	OPEN BRANCH FROM BUS 314265 TO BUS 314584 CKT 1 115.00 /* 3FIVEFORKSDP115.00 - 3LITTLTN
	OPEN BRANCH FROM BUS 314265 TO BUS 314673 CKT 1 3PALMERSPRNG115.00 /* 3FIVEFORKSDP115.00 -
	OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 115.00 /* 3CAROLNA 115.00 - 3L GASTN
	OPEN BRANCH FROM BUS 314584 TO BUS 314585 CKT 1 115.00 /* 3LITTLTN 115.00 - 3L GASTN
	OPEN BUS 314265 /* ISLAND: 3FIVEFORKSDP115.00
	OPEN BUS 314584 /* ISLAND: 3LITTLTN 115.00
	OPEN BUS 314585 /* ISLAND: 3L GASTN 115.00
	END

Contingency Name	Contingency Definition
DVP_P7-1: LN 22-90-B	CONTINGENCY 'DVP_P7-1: LN 22-90-B'
	OPEN BRANCH FROM BUS 925780 TO BUS 314702 CKT 1 115.00 /* AC1-054 TAP 115.00 - 3KERR
	OPEN BRANCH FROM BUS 314265 TO BUS 314584 CKT 1 115.00 /* 3FIVEFORKSDP115.00 - 3LITTLTN
	OPEN BRANCH FROM BUS 314265 TO BUS 314673 CKT 1 3PALMERSPRNG115.00 /* 3FIVEFORKSDP115.00 -
	OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 115.00 /* 3CAROLNA 115.00 - 3L GASTN
	OPEN BRANCH FROM BUS 314584 TO BUS 314585 CKT 1 115.00 /* 3LITTLTN 115.00 - 3L GASTN
	OPEN BUS 314265 /* ISLAND: 3FIVEFORKSDP115.00
	OPEN BUS 314584 /* ISLAND: 3LITTLTN 115.00
	OPEN BUS 314585 /* ISLAND: 3L GASTN 115.00
	END

17.2 Appendix 1

(DVP - DVP) The 6CLUBHSE 230/115 kV transformer (from bus 314562 to bus 314563 ckt 1) loads from 124.41% to 125.37% (AC power flow) of its load dump rating (209 MVA) for the line fault with failed breaker contingency outage of 'DVP_P4-6: CAROLIN T122'. This project contributes approximately 2.36 MW to the thermal violation.

```
CONTINGENCY 'DVP_P4-6: CAROLIN T122'          /* CAROLINA 115 KV
OPEN BRANCH FROM BUS 314559 TO BUS 315126 CKT 1 /* 3CAROLNA 115.00 - 1ROARAP2 14.400
OPEN BRANCH FROM BUS 314559 TO BUS 315128 CKT 1 /* 3CAROLNA 115.00 - 1ROARAP4 14.400
OPEN BUS 315126                               /* ISLAND: 1ROARAP2 14.400
OPEN BUS 315128                               /* ISLAND: 1ROARAP4 14.400
OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00
OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00
OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00
OPEN BRANCH FROM BUS 314559 TO BUS 314578 CKT 1 /* 3CAROLNA 115.00 - 3HORNRTN 115.00
OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00
OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00
OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00
OPEN BUS 314559                               /* 3CAROLNA 115.00 KV
END
```

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315158	1KERR 1	0.25
315159	1KERR 2	1.
315160	1KERR 3	0.99
315161	1KERR 4	0.99
315162	1KERR 5	0.99
315163	1KERR 6	0.99
315164	1KERR 7	0.99
314572	3EMPORIA	1.99
314704	3LAWRENC	1.13
936261	AD2-033 C	10.79
936262	AD2-033 E	7.2
936361	AD2-046 C O1	7.07
936362	AD2-046 E O1	3.25
936481	AD2-063 C O1	13.81
936482	AD2-063 E O1	9.21
938371	AE1-056 C	3.53
938372	AE1-056 E	1.93
939181	AE1-148 C O1	7.09
939182	AE1-148 E O1	4.73
940241	AE2-006	0.38
940661	AE2-053 O1	2.63
942451	AE2-258	2.36
LTF	CARR	0.03

LTF	CBM-S1	0.66
LTF	CBM-S2	0.51
LTF	CBM-W1	0.86
LTF	CBM-W2	4.48
LTF	CIN	0.39
LTF	CPL	0.21
LTF	G-007	0.1
LTF	IPL	0.25
LTF	LGEE	0.11
LTF	MEC	0.78
LTF	MECS	0.42
LTF	O-066	0.63
LTF	RENSSELAER	0.02
LTF	WEC	0.1
931231	AB1-173 C	10.72
931232	AB1-173 E	5.
931241	AB1-173AC	10.72
931242	AB1-173AE	5.
923911	AB2-031 C O1	10.64
923912	AB2-031 E O1	5.24
923991	AB2-040 C O1	34.94
923992	AB2-040 E O1	28.59
924021	AB2-043 C O1	3.16
924022	AB2-043 E O1	2.82
924161	AB2-060 C O1	8.96
924162	AB2-060 E O1	4.22
924301	AB2-077 C O1	1.94
924302	AB2-077 E O1	1.29
924311	AB2-078 C O1	1.94
924312	AB2-078 E O1	1.29
924321	AB2-079 C O1	1.94
924322	AB2-079 E O1	1.29
925171	AB2-174 C O1	33.35
925172	AB2-174 E O1	30.18
925611	AC1-036 C	0.75
925612	AC1-036 E	0.68
925781	AC1-054 C O1	5.45
925782	AC1-054 E O1	2.51

17.3 Appendix 5

(DVP - CPLE) The 3KERR-3GW KING TAP 115 kV line (from bus 314702 to bus 304102 ckt 1) loads from 131.64% to 133.0% (AC power flow) of its emergency rating (199 MVA) for the tower line contingency outage of 'DVP_P7-1: LN 22-90-A'. This project contributes approximately 2.74 MW to the thermal violation.

CONTINGENCY 'DVP_P7-1: LN 22-90-A'

```

OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1      /* 3CAROLNA 115.00 - 3EATON F 115.00
OPEN BRANCH FROM BUS 314571 TO BUS 925780 CKT 1      /* 3EATON F 115.00 - AC1-054 TAP 115.00
OPEN BUS 314571                                       /* ISLAND: 3EATON F 115.00
OPEN BRANCH FROM BUS 314265 TO BUS 314584 CKT 1      /* 3FIVEFORKSDP115.00 - 3LITTLTN 115.00
OPEN BRANCH FROM BUS 314265 TO BUS 314673 CKT 1      /* 3FIVEFORKSDP115.00 - 3PALMERSPRNG115.00
OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1      /* 3CAROLNA 115.00 - 3L GASTN 115.00
OPEN BRANCH FROM BUS 314584 TO BUS 314585 CKT 1      /* 3LITTLTN 115.00 - 3L GASTN 115.00
OPEN BUS 314265                                       /* ISLAND: 3FIVEFORKSDP115.00
OPEN BUS 314584                                       /* ISLAND: 3LITTLTN 115.00
OPEN BUS 314585                                       /* ISLAND: 3L GASTN 115.00
END

```

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315158	1KERR 1	0.71
315159	1KERR 2	2.88
315160	1KERR 3	2.83
315161	1KERR 4	2.83
315162	1KERR 5	2.83
315163	1KERR 6	2.83
315164	1KERR 7	2.83
936261	AD2-033 C	12.6
936262	AD2-033 E	8.4
936361	AD2-046 C O1	17.92
936362	AD2-046 E O1	8.24
936481	AD2-063 C O1	16.09
936482	AD2-063 E O1	10.73
938371	AE1-056 C	4.15
938372	AE1-056 E	2.27
939181	AE1-148 C O1	17.24
939182	AE1-148 E O1	11.5
940241	AE2-006	0.45
940661	AE2-053 O1	6.39
942451	AE2-258	2.74
942461	AE2-259 C O1	4.44
942462	AE2-259 E O1	2.96
LTF	BLUEG	1.92
LTF	CALDERWOOD	0.42
LTF	CANNELTON	0.13
LTF	CATAWBA	0.42

<i>LTF</i>	<i>CBM-N</i>	<i>0.02</i>
<i>LTF</i>	<i>CHEOAH</i>	<i>0.39</i>
<i>LTF</i>	<i>CHILHOWEE</i>	<i>0.13</i>
<i>LTF</i>	<i>COFFEEN</i>	<i>0.23</i>
<i>LTF</i>	<i>COTTONWOOD</i>	<i>1.37</i>
<i>LTF</i>	<i>DUCKCREEK</i>	<i>0.46</i>
<i>LTF</i>	<i>EDWARDS</i>	<i>0.21</i>
<i>LTF</i>	<i>FARMERCITY</i>	<i>0.17</i>
<i>LTF</i>	<i>G-007A</i>	<i>0.13</i>
<i>LTF</i>	<i>GIBSON</i>	<i>0.08</i>
<i>LTF</i>	<i>HAMLET</i>	<i>0.81</i>
<i>LTF</i>	<i>NEWTON</i>	<i>0.59</i>
<i>LTF</i>	<i>NYISO</i>	<i>0.09</i>
<i>LTF</i>	<i>PRAIRIE</i>	<i>1.25</i>
<i>LTF</i>	<i>SANTEETLA</i>	<i>0.12</i>
<i>LTF</i>	<i>SMITHLAND</i>	<i>0.11</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.24</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.21</i>
<i>LTF</i>	<i>TVA</i>	<i>1.15</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>0.58</i>
<i>LTF</i>	<i>VFT</i>	<i>0.35</i>
<i>924021</i>	<i>AB2-043 C O1</i>	<i>3.67</i>
<i>924022</i>	<i>AB2-043 E O1</i>	<i>3.28</i>
<i>924161</i>	<i>AB2-060 C O1</i>	<i>10.44</i>
<i>924162</i>	<i>AB2-060 E O1</i>	<i>4.91</i>
<i>924301</i>	<i>AB2-077 C O1</i>	<i>2.37</i>
<i>924302</i>	<i>AB2-077 E O1</i>	<i>1.58</i>
<i>924311</i>	<i>AB2-078 C O1</i>	<i>2.37</i>
<i>924312</i>	<i>AB2-078 E O1</i>	<i>1.58</i>
<i>924321</i>	<i>AB2-079 C O1</i>	<i>2.37</i>
<i>924322</i>	<i>AB2-079 E O1</i>	<i>1.58</i>
<i>925611</i>	<i>AC1-036 C</i>	<i>0.88</i>
<i>925612</i>	<i>AC1-036 E</i>	<i>0.79</i>
<i>925781</i>	<i>AC1-054 C O1</i>	<i>15.6</i>
<i>925782</i>	<i>AC1-054 E O1</i>	<i>7.19</i>
<i>926271</i>	<i>AC1-105 C O1</i>	<i>2.87</i>
<i>926272</i>	<i>AC1-105 E O1</i>	<i>1.43</i>

Affected Systems

18 Affected Systems

18.1 Duke Energy Progress

Potential Duke Energy Progress impact on the 115 kV line 45 from Kerr Dam to GW King Tap.

Short Circuit

19 Short Circuit

The following Breakers are overdutied:

None

Stability Analysis

20 Stability Analysis

Not required for this project.

Attachment 1

System Configuration

